ASSIGNMENT QUESTIONS ON GROUP BY

1. WAQTD number of employees working in each department excluding

   PRESIDENT.

```
  1  SELECT count(*),deptno
  2  FROM EMP
  3  WHERE job not in 'PRESIDENT'
  4* GROUP BY deptno
SQL> /

  COUNT(*)     DEPTNO
---------- ----------
         6         30
         5         20
         2         10
```

2. WAQTD total salary needed to pay all the employees in each job.

```
  1  SELECT sum(sal),job
  2  FROM EMP
  3* GROUP BY job
SQL> /

  SUM(SAL) JOB
---------- ---------
      4150 CLERK
      5600 SALESMAN
      5000 PRESIDENT
      8275 MANAGER
      6000 ANALYST
```

3. WAQTD number of employees working as MANAGER in each

department.

```
  1  SELECT count(*),deptno
  2  FROM EMP
  3  WHERE job = 'MANAGER'
  4* GROUP BY deptno
SQL> /

  COUNT(*)     DEPTNO
---------- ----------
         1         30
         1         20
         1         10
```

4. WAQTD average salary needed to pay all the employees in each

department excluding deptno 20.

```
  1  SELECT avg(sal),deptno
  2  FROM EMP
  3  WHERE deptno not in 20
  4* GROUP BY deptno
SQL> /

  AVG(SAL)     DEPTNO
---------- ----------
1566.66667         30
2916.66667         10
```

5. WAQTD number of employees having character 'A' in their names in each job.

```
  1  SELECT count(*),job
  2  FROM EMP
  3  WHERE ename like '%A%'
  4* GROUP BY job
SQL> /

  COUNT(*) JOB
---------- ---------
         3 SALESMAN
         2 CLERK
         2 MANAGER
```

6. WAQTD number of employees and average salary needed to pay the employees whose salary is greater than 2000 in each department.

```
  1  SELECT count(*),avg(sal),deptno
  2  FROM EMP
  3  WHERE sal > 2000
  4* GROUP BY deptno
SQL> /

  COUNT(*)   AVG(SAL)     DEPTNO
---------- ---------- ----------
         1       2850         30
         3 2991.66667         20
         2       3725         10
```

7. WAQTD total salary needed to pay and number of SALESMEN in each

department.

```
  1  SELECT sum(sal),count('SALESMAN'),deptno
  2  FROM EMP
  3* GROUP BY deptno
SQL> /

  SUM(SAL) COUNT('SALESMAN')     DEPTNO
---------- ---------------- ----------
      9400                6         30
     10875                5         20
      8750                3         10
```

8. WAQTD number of employees with their maximum salaries in each job.

```
  1  SELECT count(*),max(sal),job
  2  FROM EMP
  3* GROUP BY job
SQL> /

  COUNT(*)   MAX(SAL) JOB
---------- ---------- ---------
         4       1300 CLERK
         4       1600 SALESMAN
         1       5000 PRESIDENT
         3       2975 MANAGER
         2       3000 ANALYST
```

9. WAQTD maximum salary given to an employee working in each department.

```
  1  SELECT max(sal),deptno
  2  FROM EMP
  3* GROUP BY deptno
SQL> /

  MAX(SAL)     DEPTNO
---------- ----------
      2850         30
      3000         20
      5000         10
```

10.WAQTD number of times each salary is present in the employee table.

```
  1  SELECT count(sal),sal
  2  FROM EMP
  3* GROUP BY sal
SQL> /

COUNT(SAL)        SAL
---------- ----------
         1       2450
         1       5000
         1       1300
         2       1250
         1       2850
         1       2975
         1       1100
         2       3000
         1        800
         1       1600
         1       1500
         1        950

12 rows selected.
```

ASSIGNMENT QUESTIONS ON HAVING CLAUSE

1. WAQTD department number and number of employees working in each department if there are at least 2 CLERKS in each department.

```
  1  select deptno,count(*)
  2  from emp
  3  where job = 'CLERK'
  4  group by deptno
  5* having count(*) >= 2
SQL> /

    DEPTNO   COUNT(*)
---------- ----------
        20          2
```

2. WAQTD department number and total salary needed to pay all employees if there are at least 4 employees in each department.

```
  1  select deptno,sum(sal)
  2  from emp
  3  group by deptno
  4* having count(*) >= 4
SQL> /

    DEPTNO   SUM(SAL)
---------- ----------
        30       9400
        20      10875
```

3. WAQTD number of employees earning salary more than 1200 in each job, and the total salary needed to pay them must exceed 3800.

```
  1  select count(*),job,sum(sal)
  2  from emp
  3  where sal > 1200
  4  group by job
  5* having sum(sal) > 3800
SQL> /

  COUNT(*) JOB         SUM(SAL)
---------- --------- ----------
         4 SALESMAN        5600
         1 PRESIDENT       5000
         3 MANAGER         8275
         2 ANALYST         6000
```

4. WAQTD department number and number of employees working only if there are exactly 2 MANAGERS in each department.

```
  1  select deptno,count(*)
  2  from emp
  3  group by deptno
  4* having count('MANAGER') = 2
SQL> /

no rows selected

SQL> ed;
Wrote file afiedt.buf

  1  select deptno,count(*)
  2  from emp
  3  where job = 'MANAGER'
  4  group by deptno
  5* having count(*) = 2
SQL> /

no rows selected
```

5. WAQTD job and maximum salary of employees in each job if max salary exceeds 2600.

```
  1  select job,max(sal)
  2  from emp
  3  group by job
  4* having max(sal) > 2600
SQL> /

JOB         MAX(SAL)
--------- ----------
PRESIDENT       5000
MANAGER         2975
ANALYST         3000
```

6. WAQTD the salaries which are repeated in the employee table.

```
  1  select sal
  2  from emp
  3  group by sal
  4* having count(*) > 1
SQL> /

       SAL
----------
      1250
      3000
```

7. WAQTD the hire dates which are duplicated in the employee table.

```
  1  select hiredate
  2  from emp
  3  group by hiredate
  4* having count(*) > 1
SQL> /

HIREDATE
---------
03-DEC-81
```

8. WAQTD average salary of each department if average salary is less than 3000.

```
  1  select avg(sal)
  2  from emp
  3  group by deptno
  4* having avg(sal) < 3000
SQL> /

  AVG(SAL)
----------
1566.66667
      2175
2916.66667
```

9. WAQTD department number if there are at least 3 employees in each department whose name has character 'A' or 'S'.

```
  1  select deptno
  2  from emp
  3  where ename like '%A%' or ename like '%S%'
  4  group by deptno
  5* having count(*) >= 3
SQL> /

    DEPTNO
----------
        30
        20
```

10.WAQTD minimum and maximum salaries of each job if minimum salary is more than 1000 and maximum salary is less than 5000.

```
  1  select min(sal),max(sal),job
  2  from emp
  3  group by job
  4* having min(sal) > 1000 and max(sal) < 5000
SQL> /

  MIN(SAL)   MAX(SAL) JOB
---------- ---------- ---------
      1250       1600 SALESMAN
      2450       2975 MANAGER
      3000       3000 ANALYST
```